

西南财经大学本科期末考试试题册（B）

（2024—2025学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（Java SE）(英) 命题教师：王磊

适用对象（年级专业）：2024级特拉华信息管理专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[☒]
- 2、本套试题共 2 道大题，共 6 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 课本[☒] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 2. 拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 4. 所有答案均需填写在答题纸上，答在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. Which of the following is not an OOP concept in Java?

- A. Polymorphism
- B. Inheritance
- C. Compilation
- D. Encapsulation

2. If you attempt to add a **long**, an **int**, and a **double** variable, the result will be a _____ value.

- A. int
- B. long
- C. float
- D. double

3. Assuming **m** and **n** are both declared as integer variables with $m = 26$ and $n = 4$, which Java expression has a value of 7? _____

- A. $m / n + 3$
- B. $m \% n + 5$
- C. $m - 10 * n / 3$
- D. $m / 5 + n * 2$

4. What will be the output of the following Java program? _____

```
class Test
{
    public static void main(String args[])
    {
```

```

        int x;
        x = 5;
        {
            int y = 6;
            System.out.print(x + " " + y);
        }
        System.out.println(x + " " + y);
    }
}

```

- A. Compilation error
- B. Runtime error
- C. 5 6 5 6
- D. 5 6 5

5. What is the output from the following Java program fragment? _____

```

String A = "";
String B = "";
int m = 5;
for (m = -1; m < 2; m++) ;
    A = A + "*";
for (m = 3; m <= 5; m++)
    B = B + "#";
System.out.print(A);
System.out.println(B);

```

- A. *###
- B. ***###
- C. *****##
- D. ***##

6. What will be the output of the following Java Program? _____

```

class Euquality{
    int x;
    int y;
    boolean isequal(){
        return (x==y);
    }
}

```

```

    }
}

class Test{
    public static void main(String args[]){
        Euqality obj = new Euqality();
        obj.x=5;
        obj.y=5;
        System.out.println(obj.isequal() );
    }
}

```

A. Runtime error.

B. 0

C. true

D. false

7. Which of these statements about constructor is correct?_____

A. A constructor must have a return statement

B. All the constructors in a class must have same name

C. There must be exactly one constructor defined for a class

D. A constructor must be modified with public in the class

8. What type of relationship exists between someMeth in classes A and someMeth in class B?_____

```

class A {
    private void someMeth() {
        System.out.println( "from class A" );
    }
}

class B extends A {
    public void someMeth( String x ) {
        System.out.println( "from class B: " + x );
    }
}

```

```
}  
}
```

- A. method overriding
- B. method overloading
- C. both method overriding and method overloading
- D. neither method overriding nor method overloading

Animal is a class describes all land animals. **Mammal** is class describes all the land animals grown up with milk. **Bird** is a class describes all birds with wings. **LandBird** is a class describes all the birds only walking. Read the following code, and answer the next 2 questions.

```
class Animal{  
    String name;  
    int age;  
    public String toString(){  
        return "This is a land animal";  
    }  
}  
  
class Mammal extends Animal{  
    int size;  
    int weight;  
    public String toString(){  
        return "This is a mammal";  
    }  
}  
  
class Bird extends Animal{  
    int size;  
    public String toString(){  
        return "This is a bird";  
    }  
}  
  
class LandBird extends Bird{  
    int lengthOfWings;  
    public String toString(){
```

```

        return "This is a land bird";
    }
}
class Test{
    public static void main(String args[ ]){
        }
    }
}

```

9. In the main method, which one is the output of below statements:_____

Animal b= new LandBird();

System.out.println(b);

- | | |
|--------------------------|---------------------|
| A. This is a land bird | B. This is a bird |
| C. This is a land animal | D. This is a mammal |

10. Suppose that we have defined **A**, **M**, **B**, **L** to be the references of an **Animal** object, a **Mammal** object, a **Bird** object and a **LandBird** object respectively in above main method.

Which of below assignment statements is legal?_____

I. A = M III. A = L

II. A = B IV. L = B

- A. II and IV only
- B. II and III only
- C. I, II and III only
- D. I, II and IV only

Part II: Write Programs (total 70 pts)

1. (10 pts) Write a program that prompts the user to enter an integer **n**, and then display all its factors on the console.

2. (15pts) Design the method **sumSeries()** that will return the double value of the sum of the following expression at given integer argument **n**.

$$\frac{1}{n} + \frac{2}{n-1} + \frac{3}{n-2} + \dots + \frac{n-1}{2} + \frac{n}{1}$$

Hints: If the value of n smaller than 1, it should return -1.0.

3. (20 pts) Design a **Printable** interface, which contains an abstract methods: **printAllDigit()**, which prints each digit separated by SPACE on the console.

Then, you should design a **Digits** class, which describes information of integer. It contains:

- A member variable named **value** denotes the value of the integer.
- A **constructor** to initialize the object.
- A **numOfDigits()** method, returns the number of digits of the integer.
- A **getDigit(int i)** method, returns the i-th digit of the integer.
- A **isOdd()** method, returns whether the integer is an odd one.
- Implemented **printAllDigit()** method, which display each digit separated by SPACE.

Here's an example of using the class:

```
Digits x = new Digits(35);
System.out.println( x + " has " + x.numOfDigits() + " digits.");
System.out.println( "The digit of " + x + " is " + x.getDigit(1) );
System.out.println(x+ " is an odd integer?" + x.isOdd() );
x.printAllDigit();
```

4. (25 pts) Please design a Cashier program for a fruit store, which may run like below. When the user inputs the weight (kg) of three types of fruits, and click the "Calculate" button, it will display the total amount (yuan) of the fruits purchased.

Requirements:

- It should have suitable Java FX GUI components and layout.
- It should contain at least one event handling process.

Type	Price(yuan)	Weight(kg)
Apple	2.5 y/kg	4
Orange	1.5 y/kg	2
Banana	1.25 y/kg	0

Total Amount: 13.0

Calculate Reset